

7.4 Client Programming

Database design and DBMS administration usually take far less time than developing the client code, be it for a web interface, for an app, or for business classes that are located between the DBMS and the actual client code. Naturally, AI tools also help with coding in this case.

In our experience, an AI chat is only useful at the design stage, that is, when you select a suitable database library or develop and test initial functions. As a result, it's much more efficient to edit the code for detailed problems directly using an AI-supported editor. All you need to do is use a function name like `getRecentOrders`, and GitHub Copilot or another AI tool will suggest suitable code. The more comparable functions or methods are available in the code base as a guide, the better this will work.

We'll refrain from printing longer code examples in this section, but merely provide a few suggestions for prompts. A basic prerequisite for meaningful answers is that the AI tool has access to the database schema. However, we've found that many AI tools are unable to extract column names correctly even with the schema. Sometimes, it helps to specify only the relevant tables in the prompt in the form of `CREATE-TABLE` commands.

Prompt: PHP/MySQL/MySQLi: I need an alphabetical list of all customers who have placed an order in the past six months. The result should be displayed in a six-column HTML table. Each result is a link to a detail page `customer?id=xxx`.

Prompt: Node.js/MySQL: I want to write a function that returns the cus-

`customers / customers since yyyy-mm-dd.`

Prompt: Python/PyMySQL: I would like to send a promotional offer to all customers who haven't placed an order in more than six months. Can you

provide a script?

Prompt: PHP/MySQLi: How can I bundle multiple SQL commands into one transaction?

Prompt: C#: Which library should I use to access an SQLite database?

Prompt: Bash: I have a list of MariaDB databases in `mydbs.txt`. I need to do a schema update on all databases. The necessary ALTER TABLE commands are in `v32.sql`. Can you provide a script?